

# YASHIKA SINGH

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## Summary

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Innovative and impact-driven AI/ML Engineer and Data Analyst with 3+ years of experience building end-to-end AI solutions and data-driven products. Recognized for leading the design and deployment of a video-based AI interviewer, delivering human-like, conversational AI experiences adopted by paying users. Uniquely skilled at bridging machine learning, generative AI, LLMs, and data engineering to build solutions that feel both intelligent and user-centered. Proven expertise in Python, SQL, Azure Data Factory, Airflow, and Power BI, with hands-on experience transforming large-scale pharmaceutical data to drive marketing and operational insights in the healthcare industry. Holds a filed patent for a novel authentication system, reflecting a strong foundation in problem-solving and originality.

## Education

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**Yeshiva University, Katz School of Science and Health**, New York, NY

Master of Science in Artificial Intelligence, GPA: 3.8/4.0

May 2025

**Galgotias College of Engineering And Technology**, Greater Noida, India

Bachelor of Technology in Computer Science, CGPA: 3.5/4.0

Aug 2021

## Work Experience

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**AI/ML Engineer Intern, Product Manager Accelerator, Boston, MA (Remote)**

Jan 2025 - May 2025

- Designed and deployed AI-driven interview simulation workflows leveraging LLMs (OpenAI, Claude) to generate context-aware product management and behavioral assessments. Scheduled for market release in July 2025, with paid customers onboarded.
- Applied NLP techniques to develop dynamic prompt engineering, follow-up logic, and structured candidate evaluation using the CIRCLE framework.
- Collaborated with cross-functional teams to operationalize models, migrate logic to production, and ensure scalability and real-time response performance.
- Led research on multimodal AI interfaces, evaluating video avatar platforms and advanced TTS models to define the next cohort's roadmap for a more human-like, voice-and-visual interview experience.

**Data Scientist, Axtria, Noida, India**

Apr 2022 - Dec 2023

- Developed a K-Means clustering model using pharmaceutical engagement data to segment HCPs (Healthcare Professionals) based on behavioral patterns, enabling stakeholders to optimize marketing strategies, personalize outreach, and enhance overall HCP engagement.
- Developed and maintained ETL pipelines using Matillion, SQL and Python.
- Automated complex data processing workflows using DAGs in Apache Airflow, reducing manual intervention and enabling autonomous operations in dynamic environments.
- Integrated MLOps best practices with CI/CD pipelines to automate AI model training, deployment and monitoring on Databricks, ensuring scalability and reliability.

**Data Analyst, Axtria, Noida, India**

Aug 2021 - Apr 2022

- Optimized automated pipelines in Azure Data Factory and Amazon Apache Airflow, reducing data processing errors and improving overall efficiency.
- Supported in maintenance and advancement of the company's product Axtria Datamax, a cloud-based data management product.
- Created interactive dashboards and data visualizations using Power BI and Tableau to deliver insights and support decision-making for internal stakeholders.

## Patent

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**Patent:** PEHCHAAN - An Authentication System and Methods for Social Media Accounts (Filed on Nov 8, 2020, Application No. 202011048750) for effective reduction of fake social media accounts.

## Projects

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**Automatic Teaching via Vision-Language RAG (Ongoing):** Develop a Vision-Language RAG model integrating advanced AI, visual data, and textual information, leveraging open-source models like Whisper and Llama 3.2 11B Vision-Instruct Model to enhance interactive learning and enrich the educational experience.

**Drowsiness Detection for Drivers:** Built a CNN-based, real-time system using TensorFlow to detect driver fatigue and improve road safety with validation accuracy around 95%.

**Detection of Cardiomegaly in Dogs:** Created a deep learning model using custom CNN for X-ray analysis and automating veterinary diagnostics with an accuracy of 76.75%, surpassing the benchmark set by VGG16 (75%).

## Skills and Certifications

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**Programming Languages:** Python, C, C++, JavaScript, SQL

**Developer Tools/Cloud:** Amazon Apache Airflow (DAGs), Azure Data Factory, AWS, Google Cloud Platform (GCP), VS Code, Git, TensorFlow, Streamlit, Tableau, Microsoft Excel, PowerBI, PyTorch, Computer Vision, Anaconda, Jupyter, Google Colab, Pandas, Scikit-learn, Matplotlib, Data Pipelines, Databricks

**Skills:** Data Analysis, Machine Learning, Deep Learning, Natural Language Processing (NLP), Convolutional Neural Networks (CNN), Automatic Speech Recognition (ASR), Advanced Analytics, Predictive Modeling, Statistics, Mathematics, Exploratory Data Analysis (EDA), Whisper, Large Language Models (LLM), Prompt Engineering, Generative AI, Retrieval-Augmented Generation (RAG), MLOps, CI/CD Pipelines, MySQL, PostgreSQL, Extract-Transform-Load (ETL), HTML5, CSS, Technical Writing and Documentation

**Trainings:** Advanced Certificate Program in Data Science - upGrad, Machine Learning A-Z™: Hands-on Python in Data Science - Udemy, Deep Learning Specialization - Coursera, Python Certificate - HackerRank, C++ - Coding Ninjas, JavaScript Essentials - Udemy, 50 Days of DSA Python Data Structures Algorithms LEETCODE

**Soft Skills:** Problem-Solving, Critical Thinking, Adaptability, Collaboration, Communication, Leadership, Attention to Detail, Time Management, Innovation